

This is a part of SFQED-Loops script collection
developed for calculating loop and tree-level processes in
Strong-Field Quantum Electrodynamics.
The scripts are available on <https://github.com/ArsenyMironov/SFQED-Loops>

If you use this script in your research, please, consider
citing our papers:

- A. A. Mironov, S. Meuren, and A. M. Fedotov, PRD 102, 053005 (2020); arXiv:2003.06909
<https://doi.org/10.1103/PhysRevD.102.053005>
- A. A. Mironov, A. M. Fedotov, 105 (3), 033005 (2022); arXiv:2109.00634
<https://doi.org/10.1103/PhysRevD.105.033005>
- Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)

If you have any questions, please, don't hesitate to contact:
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$$\text{Solid line with shaded circle} = \text{Wavy line} + \text{Wavy line with 1PI loop} + \text{Wavy line with 2PI loops} + \dots$$

```
In[1]:= NotebookEvaluate[NotebookOpen[NotebookDirectory[] <> "definitions.nb"]]
```

FeynCalc 10.1.0 (stable version). For help, use the
online documentation, visit the forum and have a look at the supplied
examples. The PDF-version of the manual can be downloaded [here](#).

If you use FeynCalc in your research, please
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*Please keep in mind that the proper academic attribution
of our work is crucial to ensure the future development of this package!*

Exact photon propagator in momentum representation

$$D_{\mu\nu}^c(l) =$$

$$D_0(l^2, \chi_l) g_{\mu\nu} + D_1(l^2, \chi_l) \epsilon_{\mu}^{(1)}(l) \epsilon_{\nu}^{(1)}(l) + D_2(l^2, \chi_l) \epsilon_{\mu}^{(2)}(l) \epsilon_{\nu}^{(2)}(l);$$

l^{μ} – the photon propagator 4 – momentum;

$$\chi_l = \frac{e}{m^3} \sqrt{-\left(F_{\mu\nu} l^{\nu}\right)^2};$$

$$\epsilon_{\mu}^{(1)}(l) = \frac{e F_{\mu\nu} l^{\nu}}{m^3 \chi_l};$$

$$\epsilon_{\mu}^{(2)}(l) = \frac{e F_{\mu\nu}^* l^{\nu}}{m^3 \chi_l}; \quad F^{*\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\delta\lambda} F_{\delta\lambda};$$

$$\left(\epsilon^{(i)}(l)\right)^2 = -1;$$

$$D_0(l^2, \chi_l) = \frac{-i}{l^2 - l^2 \hat{\Pi}}, \quad D_{1,2}(l^2, \chi_l) = \frac{i \Pi_{1,2}}{(l^2 - l^2 \hat{\Pi})(l^2 - l^2 \hat{\Pi} - \Pi_{1,2})};$$

$$l^2 \hat{\Pi} = l^2 \hat{\Pi}(l^2, \chi_l),$$

$\Pi_{1,2} = \Pi_{1,2}(l^2, \chi_l)$ – polarization operator eigenfunctions;

Our goal : exact photon propagator in coordinate representation

$$D_{\mu\nu}^c(x) = \frac{\Lambda^{4-D}}{(2\pi)^D} \int d^D l D_{\mu\nu}^c(l) e^{-ilx};$$

Let's define photon momentum and the coordinate variables

```
In[2]:= NewMomentum["l"]
NewCoordinate["x"]
```

$$\{l^{\alpha}, l^2, k \cdot l, F l^{\alpha}, F F l^{\alpha}, F D l^{\alpha}, a \cdot l, 0, 0, 0, -a^2(k \cdot l), 0, 0, -\frac{m^6 \chi l^2}{e^2}, -\frac{m^6 \chi l^2}{e^2}, \frac{m^6 \chi l^2}{e^2}, 0, 0, 0, 0, 0, 0\}$$

$$\{x^{\alpha}, x^2, k \cdot x, a \cdot x, F x^{\alpha}, F F x^{\alpha}, F D x^{\alpha}, k \cdot x, 0, 0, 0, -a^2(k \cdot x), 0, 0, -\frac{m^2 \xi^2 (k \cdot x)^2}{e^2}, -\frac{m^2 \xi^2 (k \cdot x)^2}{e^2}, \frac{m^2 \xi^2 (k \cdot x)^2}{e^2}, 0, 0, 0, 0, 0, 0\}$$

Eigenvectors and tensor structures

We intentionally leave tensors and vectors uncontracted

```

In[4]:= e1[μ_, ν_] = e Ft[μ, ν] × lv[ν] / m^3 / χl
e2[μ_, ν_] = e FDt[μ, ν] × lv[ν] / m^3 / χl
T0[μ_, ν_] = MTD[μ, ν]
T1[μ_, ν_] = e1[μ, α] e1[ν, β]
T2[μ_, ν_] = e2[μ, α] e2[ν, β]
Contract[Contract[T1[μ, μ]] /. FieldSubstitutions]
Contract[Contract[T2[μ, μ]] /. FieldSubstitutions]

```

$$\text{Out[4]} = \frac{e l^\nu F(\mu, \nu)}{m^3 \chi l}$$

$$\text{Out[5]} = \frac{e l^\nu \text{FD}(\mu, \nu)}{m^3 \chi l}$$

$$\text{Out[6]} = g^{\mu \nu}$$

$$\text{Out[7]} = \frac{e^2 l^\alpha l^\beta F(\alpha, \mu) F(\beta, \nu)}{m^6 \chi l^2}$$

$$\text{Out[8]} = \frac{e^2 l^\alpha l^\beta \text{FD}(\alpha, \mu) \text{FD}(\beta, \nu)}{m^6 \chi l^2}$$

$$\text{Out[9]} = -1$$

$$\text{Out[10]} = -1$$

We write $D^c_{\mu\nu}$ in the following form

$$\int d^D l \left[\text{Coeff} * \text{Matrix} * \text{Exp} (i \text{ Phase}) \right],$$

where

Coeff – is a general multiplier for all terms,

Matrix – tensor part,

Phase – total phase of the expression,

We assume

$$Dk = Dk (l^2, \chi l)$$

```

In[11]:= Coeff = Δ^ (4 - D) / (2 π) ^ D
Matrix = D0 * T0[μ, ν] + D1 * T1[μ, ν] + D2 * T2[μ, ν]
Phase = -Contract[xv[α] × lv[α]]

```

$$\text{Out[11]} = (2 \pi)^{-D} \Lambda^{4-D}$$

$$\text{Out[12]} = \frac{D1 e^2 l^\alpha l^\beta F(\alpha, \mu) F(\beta, \nu)}{m^6 \chi l^2} + \frac{D2 e^2 l^\alpha l^\beta \text{FD}(\alpha, \mu) \text{FD}(\beta, \nu)}{m^6 \chi l^2} + D0 g^{\mu \nu}$$

$$\text{Out[13]} = -(l \cdot x)$$

We need to calculate the integrals of two types

$$\int d^D l D_0 (l^2, \chi l) e^{-i l x};$$

and

$$\int d^D l \, l_\alpha l_\beta D_{1,2}(l^2, x_l) e^{-i l x} = i \frac{\partial}{\partial x^\alpha} i \frac{\partial}{\partial x^\beta} \int d^D l \, D_{1,2}(l^2, x_l) e^{-i l x};$$

Symbol $d_{\alpha,\beta}$ means that we need to differentiate the expression later

Let us now change the variables

$$l_m = l_- = l_0 - l_3;$$

$$l_p = l_+ = \frac{l_0 + l_3}{2};$$

$$l_t = l_\perp = \text{transverse components of } l \text{ (in } D=4 \text{ } l_\perp = (l_1, l_2));$$

$$l^2 = l^2$$

$$l^2 = 2 l_- l_+ - l_\perp^2;$$

Proper time

$$s = x_- / 2 l_- = k x / 2 k l;$$

$$l_- = x_- / 2 s;$$

$$l_+ = (l^2 + l_\perp^2) / 2 l_- = s (l^2 + l_\perp^2) / x_- \quad \text{-- expressed via } l^2;$$

Hereinafter

$$x_m = x_- = x_0 - x_3;$$

$$x_p = x_+ = \frac{x_0 + x_3}{2};$$

$$x_t = x_\perp$$

Change of variables

$$l^\mu \rightarrow \{l_-, l_+, l_\perp\} = \left\{ \frac{x_-}{2s}, \frac{s}{x_-} (l^2 + l_\perp^2), l_\perp \right\} \rightarrow \{s, l^2, l_\perp\}$$

New integration measure

$$d^D l \dots = \frac{ds}{2|s|} dl^2 d^{D-2} l_\perp$$

In[14]:= (*Checking Jacobian*)

D[{xm/2/s, s (lv2+lt2)/xm}, {{s, lv2}}]

Jac = Abs[Det[%]]

$$\text{Out[14]} = \begin{pmatrix} -\frac{x_m}{2s^2} & 0 \\ \frac{l^2+lt2}{x_m} & \frac{s}{x_m} \end{pmatrix}$$

$$\text{Out[15]} = \frac{1}{2|s|}$$

```
In[16]:= Coeff1 = Coeff * Jac
Matrix1 = Matrix /. {lv[α_] → dα}
Phase1 = Phase;
```

$$\text{Out[16]} = \frac{2^{-D-1} \pi^{-D} \Lambda^{4-D}}{|s|}$$

$$\text{Out[17]} = \frac{D1 e^2 d_\alpha d_\beta F(\alpha, \mu) F(\beta, \nu)}{m^6 \chi l^2} + \frac{D2 e^2 d_\alpha d_\beta \text{FD}(\alpha, \mu) \text{FD}(\beta, \nu)}{m^6 \chi l^2} + D0 g^{\mu \nu}$$

Integration over

$$\int d^{D-2} l_\perp \dots$$

$$\begin{aligned} I_0 &= \int d^{D-2} l_\perp \text{Exp} \left[-I A l_\perp^2 + I (J \cdot l_\perp) \right] = \\ &= \text{Exp} \left[-I \frac{\pi}{2} \frac{D-2}{2} \right] \pi^{\frac{D-2}{2}} (\det A)^{-\frac{1}{2}} \text{Exp} \left[I \frac{1}{4} J \cdot A^{-1} \cdot J \right] \end{aligned}$$

where

$$A = s,$$

$$J = x_\perp,$$

$$\det A = s^{D-2},$$

$$A^{-1} = 1 / s$$

In effect, integration results into
multiplication of Coeff by I_0 and changing Phase

```
In[19]:= Coeff2 = Coeff1;
Matrix2 = Matrix1;
Phase2 =
Expand[Phase1 /. {Pair[Momentum[x, D], Momentum[l, D]] → xm lp + xp lm - xt lt} /.
{lm → xm / 2 / s, lp → s (lv2 + lt^2) / xm} /. {xp → (xv2 + xt^2) / 2 / xm}]
```

$$\text{Out[21]} = -l^2 s + lt^2 (-s) + lt xt - \frac{x^2}{4s} - \frac{xt^2}{4s}$$

```
In[22]:= Amatr = -Coefficient[Phase2, lt^2]
J = Coefficient[Phase2, lt]
CI0 = Exp[-I Pi / 2 (D / 2 - 1)] Pi^ (D / 2 - 1) / Amatr^ (D / 2 - 1)
```

$$\text{Out[22]} = s$$

$$\text{Out[23]} = xt$$

$$\text{Out[24]} = e^{-\frac{1}{2} i \pi \left(\frac{D}{2} - 1\right)} \pi^{\frac{D}{2} - 1} s^{1 - \frac{D}{2}}$$

```
In[25]:= Coeff3 = Coeff2 * CI0
Matrix3 = Matrix2;
Phase3 = Expand[ ( (Phase2 /. {lt -> 0}) + 1 / 4 J^2 / Amatr) ]
```

$$\text{Out[25]} = \frac{2^{-D-1} e^{-\frac{1}{2} i \pi \left(\frac{D}{2}-1\right)} \pi^{-\frac{D}{2}-1} \Lambda^{4-D} s^{1-\frac{D}{2}}}{|s|}$$

$$\text{Out[27]} = l^2 (-s) - \frac{x^2}{4 s}$$

The integrals

$$\int_{-\infty}^{\infty} \frac{ds}{2|s|} e^{-i \frac{x^2}{4s}} \int_{-\infty}^{\infty} dl^2 D_k(l^2, \chi_l) e^{-i l^2 s}$$

will remain

Let us introduce

$$J_k = J_k(s) = -i \int_{-\infty}^{\infty} dl^2 D_k(l^2, \chi_l) e^{-i l^2 s} \quad \text{-- note that this integral is dimensionless}$$

It can be shown that $J_k(s \leq 0) = 0$

```
In[28]:= Coeff4 = Simplify[Coeff3 * I, Assumptions -> {s > 0}]
Matrix4 = Matrix3 /. {D0 -> J0, D1 -> J1, D2 -> J2}
Phase4 = Phase3 /. {lv2 -> 0}
```

$$\text{Out[28]} = -2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} \Lambda^{4-D} s^{-D/2}$$

$$\text{Out[29]} = \frac{e^2 J_1 d_\alpha d_\beta F(\alpha, \mu) F(\beta, \nu)}{m^6 \chi l^2} + \frac{e^2 J_2 d_\alpha d_\beta \text{FD}(\alpha, \mu) \text{FD}(\beta, \nu)}{m^6 \chi l^2} + J_0 g^{\mu \nu}$$

$$\text{Out[30]} = -\frac{x^2}{4 s}$$

Calculation of the tensor structure

Let us perform the remaining differentiation
and contract the resulting vectors and tensors

```
In[31]:= Expand[Simplify[
  I FourDivergence[I FourDivergence[Exp[I Phase4], xv[α]], xv[β]] Exp[-I Phase4]]]
Matrix4 /. {dα dβ → %}
Matrix5 = Collect[Contract[
  Contract[%] /. FieldSubstitutions /. FieldSubstitutions], {J0, J1, J2}]
Coeff5 = Coeff4
Phase5 = Phase4
```

$$\text{Out[31]} = \frac{x^\alpha x^\beta}{4 s^2} + \frac{i g^{\alpha\beta}}{2 s}$$

$$\text{Out[32]} = \frac{e^2 J1 F(\alpha, \mu) F(\beta, \nu) \left(\frac{x^\alpha x^\beta}{4 s^2} + \frac{i g^{\alpha\beta}}{2 s} \right)}{m^6 \chi l^2} + \frac{e^2 J2 \text{FD}(\alpha, \mu) \text{FD}(\beta, \nu) \left(\frac{x^\alpha x^\beta}{4 s^2} + \frac{i g^{\alpha\beta}}{2 s} \right)}{m^6 \chi l^2} + J0 g^{\mu\nu}$$

$$\text{Out[33]} = J2 \left(\frac{e^2 \text{FD}x^\mu \text{FD}x^\nu}{4 m^6 s^2 \chi l^2} - \frac{i e^2 \text{FF}(\mu, \nu)}{2 m^6 s \chi l^2} \right) + J1 \left(\frac{e^2 \text{F}x^\mu \text{F}x^\nu}{4 m^6 s^2 \chi l^2} - \frac{i e^2 \text{FF}(\mu, \nu)}{2 m^6 s \chi l^2} \right) + J0 g^{\mu\nu}$$

$$\text{Out[34]} = -2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} \Lambda^{4-D} s^{-D/2}$$

$$\text{Out[35]} = -\frac{x^2}{4 s}$$

Change to the dimensionless proper time

```
In[36]:= Coeff6 = Simplify[Coeff5 / m^2 /. {s → t / m^2}, Assumptions → {m > 0}]
Matrix6 = Matrix5 /. {χ l → ξ k l / m^2} /. {k l → k x / 2 / s} /. {k x → φ} /.
  {FFt[μ, ν] → -av2 kv[μ] × kv[ν]} /. {av2 → -m^2 ξ^2 / e^2} /. {s → t / m^2}
Phase6 = Phase5 /. {s → t / m^2}
```

$$\text{Out[36]} = -2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} \Lambda^{4-D} m^{D-2} t^{-D/2}$$

$$\text{Out[37]} = J2 \left(\frac{e^2 \text{FD}x^\mu \text{FD}x^\nu}{m^2 \xi^2 \phi^2} - \frac{2 i t k^\mu k^\nu}{m^2 \phi^2} \right) + J1 \left(\frac{e^2 \text{F}x^\mu \text{F}x^\nu}{m^2 \xi^2 \phi^2} - \frac{2 i t k^\mu k^\nu}{m^2 \phi^2} \right) + J0 g^{\mu\nu}$$

$$\text{Out[38]} = -\frac{m^2 x^2}{4 t}$$

Final result

$$\begin{aligned}
D_{\mu\nu}^c(x) &= \frac{\Lambda^{4-D}}{(2\pi)^D} \int d^D l D_{\mu\nu}^c(l) e^{-ilx} = \\
&= \text{Exp} \left[-i \frac{\pi}{2} \left(\frac{D}{2} - 2 \right) \right] \frac{\Lambda^{4-D}}{2^{D+1} \pi^{D/2+1}} \\
&\quad \int_0^\infty \frac{ds}{s^{D/2}} e^{-i \frac{x^2}{4s}} \left\{ g_{\mu\nu} J_0(s) - i \frac{1}{2s m^6 \chi_1^2} \left(e^2 (F^2)_{\mu\nu} + i \frac{1}{2s} e^2 F x_\mu F x_\nu \right) J_1(s) \right. \\
&\quad \left. - i \frac{1}{2s m^6 \chi_1^2} \left(e^2 (F^2)_{\mu\nu} + i \frac{1}{2s} e^2 F D x_\mu F D x_\nu \right) J_2(s) \right\} = \\
&= \text{Exp} \left[-i \frac{\pi}{2} \left(\frac{D}{2} - 2 \right) \right] \frac{1}{2^{D+1} \pi^{D/2+1}} \frac{\Lambda^{4-D}}{m^{2-D}} \\
&\quad \int_0^\infty \frac{dt}{t^{D/2}} e^{-i \frac{m^2 x^2}{4t}} \left\{ g_{\mu\nu} J_0(m^{-2} t) + \left(-2 i t \frac{k_\mu k_\nu}{m^2 \phi^2} + \frac{e^2 F x_\mu F x_\nu}{m^2 \xi^2 \phi^2} \right) J_1(m^{-2} t) \right. \\
&\quad \left. + \left(-2 i t \frac{k_\mu k_\nu}{m^2 \phi^2} + \frac{e^2 F D x_\mu F D x_\nu}{m^2 \xi^2 \phi^2} \right) J_2(m^{-2} t) \right\}; \\
J_k(s, \chi_1) &= -i \int_{-\infty}^\infty dl^2 D_k(l^2, \chi_1) e^{-il^2 s}; \\
\phi &= kx; \\
\chi_1 &= \xi k l / m^2 = \xi \phi / 2 m^2 s = \xi \phi / 2 t; \\
\xi^2 &= -e^2 a^2 / m^2; \\
F x_\mu &= F_{\mu\nu} x^\nu; \\
F D x_\mu &= F_{\mu\nu}^* x^\nu;
\end{aligned}$$